

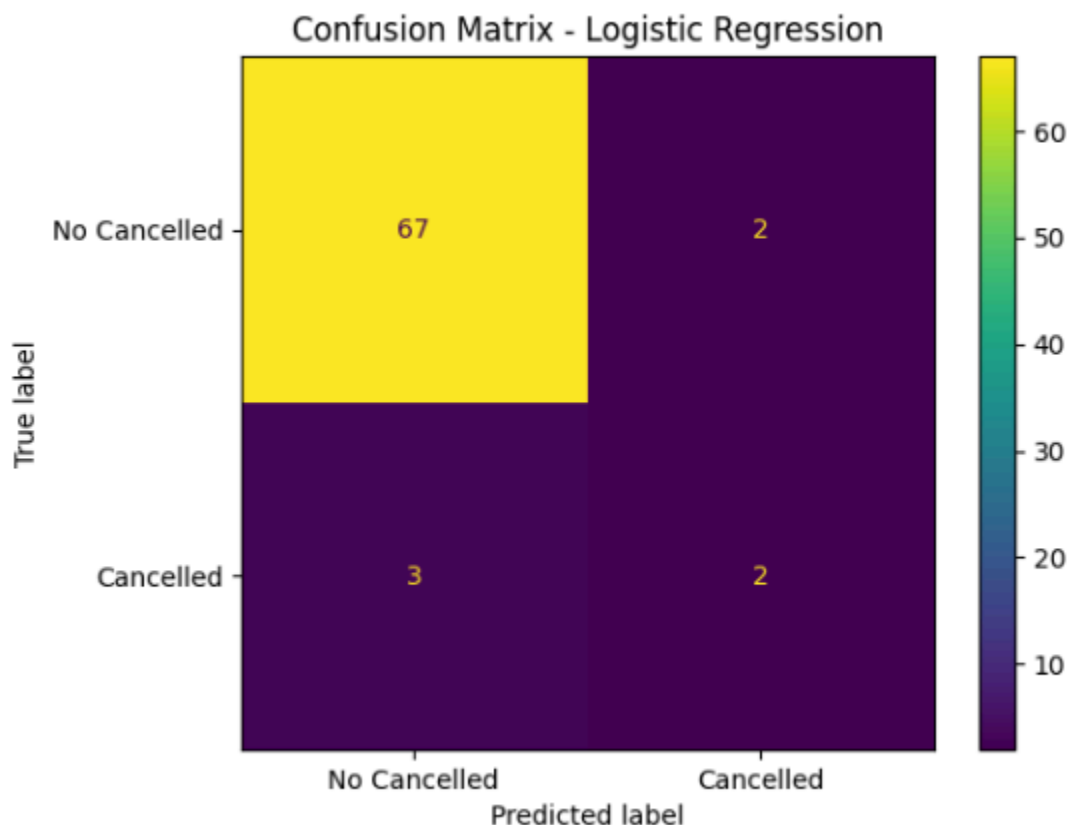
Model Evaluation Report

Hotel Booking Cancellation Prediction - Logistic Regression

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1. Confusion Matrix: Interpretation

The confusion matrix below summarises how the Logistic Regression model performed on the test set of 74 hotel booking records. Rows represent the actual (true) class; columns represent the class the model predicted.



1.1 True Negative (TN = 67)

In the context of this dataset, these are hotel guests who did not cancel their bookings, and the model correctly identified them as non-cancellers. With 67 out of 74 test observations falling into this category, the model is very good at recognising loyal, non-cancelling customers. From a business perspective, correctly identifying these guests avoids unnecessary interventions (such as unsolicited discount offers or overbooking allowances) that would waste resources or damage customer relationships.

1.2 False Positive (FP = 2)

These are guests the model wrongly flagged as likely to cancel when they would have stayed. In hotel revenue management, a false positive means the property might take unnecessary preventive action. For example, offering a discount or imposing a stricter deposit policy on a guest who never intended to cancel. With only 2 false positives, this error type is small and relatively benign in this dataset.

1.3 False Negative (FN = 3)

These are the most operationally costly errors for a hotel. The model missed 3 out of 5 actual cancellations in the test set. A missed cancellation means the hotel did not take any retention action, no reminder, no incentive, no policy change, and the room ended up vacant with lost revenue. In a real deployment, minimising false negatives (i.e., maximising recall on the cancellation class) should be a core optimisation target.

1.4 True Positive (TP = 2)

The model correctly identified 2 out of 5 actual cancellations. While any correct cancellation prediction is useful, allowing the hotel to proactively contact the guest or adjust overbooking strategies, capturing only 40% of real cancellations highlights a significant weakness in the model's ability to detect the minority class.

2. Evaluation Metrics: Results and Meaning

The following table presents all evaluation metrics produced by the classification report in the notebook. Metrics are shown per class and as weighted averages across the test set.

Classification Report:					
		precision	recall	f1-score	support
No Cancelled		0.96	0.97	0.96	69
Cancelled		0.50	0.40	0.44	5
	accuracy			0.93	74
	macro avg	0.73	0.69	0.70	74
	weighted avg	0.93	0.93	0.93	74

2.1 Accuracy (93.24%)

Accuracy measures the proportion of all predictions, both cancellations and non-cancellations, that the model got right.

The model correctly classifies 93.24% of all test records. On the surface, this appears excellent. However, this number is heavily inflated by the dominant majority class: 69 out of 74 test records are non-cancellations. A naive model that predicts "No Cancellation" for every single booking would achieve $69/74 = 93.2\%$ accuracy, essentially the same score, without ever detecting a single real cancellation.

This illustrates why accuracy alone is misleading for imbalanced classification problems.

2.2 Precision (Cancellation Class = 50%)

Precision answers: of all the bookings the model flagged as cancellations, what proportion actually cancelled?

A precision of 50% for the cancellation class means that when the model raises a cancellation alert, it is only correct half the time. The other half are guests who would not have cancelled (false positives). For a hotel's marketing or operations team, this means half of proactive retention actions would be directed at guests who did not need them.

2.3 Recall (Cancellation Class = 40%)

Recall (also called sensitivity or true positive rate) answers: of all bookings that actually cancelled, what proportion did the model catch?

A recall of 40% is critically low in this context. It means the model missed 3 out of 5 actual cancellations in the test set. For the hotel, these are guests who cancelled without the business ever being alerted, representing directly avoidable revenue losses. Improving recall on the cancellation class (even at the cost of lower precision) would be a priority in a real-world deployment.

2.4 F1-Score (Cancellation Class = 0.44)

The F1-score is the harmonic mean of Precision and Recall. It provides a single balanced metric when both false positives and false negatives carry cost.

An F1-score of 0.44 for the cancellation class indicates poor overall performance on the class that matters most to the business. The harmonic mean penalises extreme imbalances between precision and recall more strongly than a simple average would, making it a better summary metric when one class is rare.

3. Accuracy vs F1-Score: A Critical Comparison

The gap between the headline accuracy (93.24%) and the cancellation-class F1-score (0.44) is the central finding of this evaluation. The table below makes the contrast explicit:

Dimension	Accuracy (93.24%)	F1-Score - Cancelled (0.44)
What it measures	Overall correct predictions across both classes	Balance of precision & recall for the cancellation class only
Influenced by class imbalance?	Yes, dominated by the majority class (No Cancel = 93% of data)	No, computed purely within the minority class
What this model's score reveals	Misleadingly high because the model mostly predicts No Cancel	Truthfully low: the model struggles to detect actual cancellations

Appropriate use	Useful only when classes are balanced	Preferred metric for this imbalanced cancellation task
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3.1 Why Accuracy Overstates Performance

The test set contains 69 non-cancellations and only 5 cancellations (a roughly 14:1 ratio). Because the model correctly predicts the majority class almost every time (97% recall on No Cancel), accuracy stays high regardless of how poorly it handles cancellations. This is known as the accuracy paradox: a classifier can report high accuracy while being nearly useless for the business problem it was designed to solve.

3.2 Why F1-Score Is the More Honest Metric Here

The F1-score for the cancellation class cuts through the imbalance. It holds the model accountable for both its false alarms (low precision) and its missed cancellations (low recall). An F1 of 0.44 on cancellations correctly signals that the model performs only slightly above chance on the task the business actually cares about, predicting which bookings will be cancelled.

3.3 Recommendations for Improvement

To improve performance on the cancellation class, the following techniques should be considered in future iterations:

- Class rebalancing: Apply SMOTE (Synthetic Minority Over-sampling Technique) or class weighting (`class_weight='balanced'` in sklearn) to give the model more signal from cancellation examples.
- Threshold tuning: Lower the decision threshold from the default 0.5 to increase recall at the cost of precision.
- More data: The test set contains only 5 actual cancellations; a larger dataset would provide more reliable estimates of model performance.
- Alternative models: Tree-based models such as Random Forest or Gradient Boosting may better capture non-linear patterns associated with cancellations.